

Machine Learning Model Outcomes

Executive summary

Overview

The TikTok data team aims to develop a machine learning model to classify videos as either claims or opinions. Earlier analysis showed that video engagement metrics are strong indicators of claim status. The team is confident that the model will achieve the desired performance standards.

Problem

TikTok videos receive a high volume of user reports for various reasons, making it impractical for all reported videos to be reviewed by human moderators. Videos that present claims are more likely to contain content that violates the platform's terms of service. TikTok seeks a method to identify claim videos in order to prioritize them for review.

Solution

The data team developed two tree-based classification models. Both models were evaluated on a held-out validation dataset, with final selection based on the highest recall score. The chosen model was then applied to a test dataset to estimate its performance on future data.

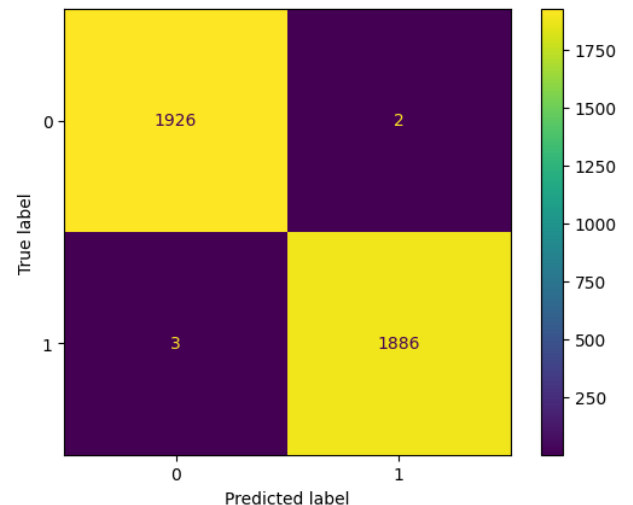
Details

Both model architectures—Random Forest (RF) and XGBoost—performed exceptionally well. The RF model achieved a higher recall score of 0.995 and was selected as the champion model.

Performance on the test holdout dataset was nearly perfect, with only five misclassifications out of 3,817 samples.

Subsequent analysis confirmed that the primary predictors were all related to video engagement, including view count, like count, share count, and download count. These features accounted for nearly all the predictive signal in the data. The results indicate that videos with higher engagement levels are much more likely to be claims. Notably, no opinion video exceeded 10,000 views.

Confusion matrix for the champion RF model on test holdout data shows only five misclassified samples out



Next Steps

As noted, the model performed exceptionally well on the test holdout dataset. Before deployment, the data team recommends further evaluation using additional subsets of user data. They also advise monitoring the distributions of video engagement metrics to ensure the model remains robust to changes in its most predictive features.